NAVANEETH AMARNATH

Research Scientist L2 at Infrrd

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PROFILE

Research Scientist specializing in agentic automation, large language models, and document intelligence, with end-to-end ownership of deployed systems and research pipelines.

WORK EXPERIENCE

Research Scientist L2 | Infrrd **Apr 2026 - Present**

- Designed and built a mortgage quality control agent from proof of concept through deployment for live post-closing review.
- Developed a harness that guides the agent through a loan represented as an environment to produce a three-class classification, complementing static workflows with autonomous processing.
- Ported an internal agentic workflow framework from AutoGen to Claude Agent SDK, adopting a harness-based architecture to support greater autonomy.

Research Scientist L1 | Infrrd **Jul 2024 - Mar 2026**

- Independently designed and developed a multi-agent framework that plans, designs, codes, and tests executable workflows from business rules; applied it to mortgage underwriting research.
- Developed a document classification pipeline using bi-encoder models.
- Built evaluation frameworks and benchmarking pipelines for ML and LLM applications.
- Created an LLM-as-a-judge feedback pipeline to iteratively refine document extraction prompts.
- Built business-data chatbots and worked on text extraction from engineering diagrams.

Research - ML Intern | Infrrd **Jun 2023 - Jul 2024**

- Fine-tuned compact language models for named entity recognition and built datasets and evaluation pipelines for ML research.
- Built retrieval-augmented generation systems for domain knowledge and few-shot example selection.
- Developed a chat application for extracting structured data from PDF documents.

PUBLICATIONS

Planning Agent Assisted Static Workflow Generation for Mortgage Underwriting Automation. S. Nakshathri, **N. Amarnath**, M. N, P. Rasamsetty and D. Kumar. IEEE COMPSAC, 2026, pp. 2392-2399.

Multi-Stage Retriever Model for Document Classification Using Fine Tuned Embedding Model. N. Amarnath, S. Tallam, P. Rasamsetty and D. Kumar. IEEE CCWC, 2025, pp. 209-218.

Infrrd.ai at SemEval-2024 Task 7: RAG-based end-to-end training to generate headlines and numbers. J. He, S. Tallam, S. Nakshathri, **N. Amarnath**, P. Kr and D. Kumar. SemEval, 2024, pp. 940-951.

EDUCATION

B.Tech, Computer Science and Engineering | Vellore Institute of Technology, Chennai | 2024
Specialization: AI and Robotics | CGPA: 8.43
Top 10, Bajaj Finserv HackRx 4.0 (2023)

SKILLS

Methods: Agentic automation, LLM applications, retrieval-augmented generation, model evaluation, prompt engineering, computer vision
Tools: Python, PyTorch, TensorFlow, LangGraph, CrewAI, Claude Agent SDK, AutoGen, LangChain, Transformers, PEFT, MongoDB, SQL

LANGUAGES

English - Hindi - Malayalam